

U.G.C. NET Exam

COMPUTER

(Solved with Expl)

1. When data and acknowledgement are sent in the same frame, this is called as

- (a) Piggy packing
- (b) Piggy backing
- (c) Back packing
- (d) Good packing

Ans : (b) Piggy backing is a bi-directional data transmission technique in the network layer (OSI model). It marks the most of the sent data frames from receiver, to emitter adding the confirmation that the data frame sent by the sender was received successfully

2. Encryption and Decryption is the responsibility of Layer.

- (a) Physical
- (b) Network
- (c) Application
- (d) Datalink

Ans : (c) The sixth layer of OSI model, responsible for translation, encryption, authentication and data compression.

The application layer of the TCP/IP model combines the functionalities of the session layer, the presentation layer, and the application layer of the OSI model. Encryption and Decryption is the responsibility of application layer (in TCP/IP model application layer + presentation layer + session layer).

3. An analog signal carries 4 bits in each signal unit. If 1000 signal units are sent per second, then band rate and bit rate of the signal are and

- (a) 4000 bauds/sec & 1000 bps
- (b) 2000 bauds/sec & 1000 bps
- (c) 1000 bauds/sec & 500 bps
- (d) 1000 bauds/sec & 4000 bps

Ans : (d) Bit rate is the number of bits per second. Baud rate is the number of signal unit per second. Baud rate is less than or equal to the bit rate.

So here baud rate is given 1000 bauds (signal) per sec
no. of bits per signal = 4

So bit rate = 4000 bits per sec

4. The VLF and LF bauds use propagation for communication.

- (a) Ground
- (b) Sky
- (c) Line of sight
- (d) Space

Ans : (a) In general there are three types of propagation
(1) Ground propagation
(2) Sky propagation
(3) Line of sight propagation

n. December, 2013

R SCIENCE

anation Paper–II)

In ground propagation, radio waves travel through the lowest portion of the atmosphere. These low-frequency signals travel in all directions from the transmitting antenna.

5. Using the RSA public key crypto system, if $p = 13$, $q = 31$ and $d = 7$, then the value of e is :

- (a) 101
- (b) 103
- (c) 105
- (d) 107

Ans : (b) $\phi(n) = (P - 1)(Q - 1) = 12 \times 30 = 360$

choose an int e such that $1 < e < \phi$ & e & ϕ are co prime
 $d \cdot e = 1 \pmod{Q}$

$7e = 1 \pmod{360} = 361 \pmod{360} = 721 \pmod{360} \Rightarrow 7e = 721 \Rightarrow e = 103$

6. FAN IN of a component A is defined as :

- (a) Number of components that call or pass control to component A.
- (b) Number of components that are called by component A.
- (c) Number of components related to component A.
- (d) Number of components dependent on component A.

Ans : (a) FANIN $\Rightarrow$ number of components that can call or pass control to component A

FAN OUT $\Rightarrow$ number of component that are called by component A

7. The relationship of data elements in a module is called

- (a) Coupling
- (b) Modularity
- (c) Cohesion
- (d) Granularity

Ans : (c) The relationship of data elements in a module is called cohesion.

Cohesion refers to the degree to which the elements of a module belong together. Thus, cohesion measures the strength of relationship between pieces of functionality within a given module. For example. in highly cohesive systems functionality is strongly related.

8. Software configuration management is the discipline for systematically controlling :

- (a) the changes due to the evolution of work products as the project proceeds
- (b) the changes due to defects (bugs) being found and then fixed
- (c) the changes due to requirement changes
- (d) All of the above

Ans: (d) In software engineering, software configuration management is the task of tracking & controlling changes in the software, part of the larger cross-disciplinary field of configuration management. SCM practices include revision control and the establishment of base line.

9. Which one of the following is not a step of requirement engineering?

- (a) Requirement elicitation
- (b) Requirement analysis
- (c) Requirement design
- (d) Requirement documentation

Ans : (c) **Requirements elicitation** is requirements gathering.

Requirement design is not a step of requirement engineering.

10. Testing of software with actual data and in actual environment is called :

- (a) Alpha testing
- (b) Beta testing
- (c) Regression testing
- (d) None of the above

Ans : (b) Testing of software with actual data and in actual-environment is called Beta testing.

11. The student marks should not be greater than 100. This is:

- (a) Integrity constraint
- (b) Referential constraint
- (c) Over-defined constraint
- (d) Feasible constraint

Ans : (a) Constraints are the condition that must be satisfied by all the tuples of the relation.

The types of constraints are follows-

- Key constraints
- Entity integrity constraints
- Referential integrity constraints

Key Constraints are the condition that must be satisfied by all the tuples of the relation.

Key constraint tells that there should be a key to identify the unique tuple.

Entity integrity constraints tells that primary key value cannot be null.

Referential integrity rule 2 says that a foreign key in table A refers to the primary key of B. Then every value of foreign key in table A must be null or be available in table B. Create table abc(name varchar (20), marks integer check marks <= 100), It will not allow to insert marks greater than 100.

12. GO BOTTOM and SKIP-3 commands are given one after another in a database file of 30 records. It shifts the control to :

- (a) 28th record (b) 27th record
- (c) 3rd record (d) 4th record

Ans : (b) Database file contain 30 records:

Go bottom then skip 3

Go to bottom (at last 30th record) then skip 3 records (i.e., 30th, 29th, and 28th). Then you are at 27th record.

13. An ER model includes

- (i) An ER diagram portraying entity types
 - (ii) Attributes for each entity type
 - (iii) Relationships among entity types
 - (iv) Semantic integrity constraints that reflects the business rules about data not captured in the ER diagram
- (a) 1, 2, 3 & 4 (b) 1 & 4
 - (c) 1, 2 & 4 (d) 1 & 3

Ans : (a) ER model includes

I. An ER diagram portraying entity types.

II. Attributes for each entity type

III. Relationships among entity types

IV. Semantic integrity constraints that reflects the business rules about data not captured in the ER diagram.

14. Based on the cardinality ratio and participation associated with a relationship type, choose either the Foreign key design, the cross referencing design or mutual referencing design.

- (a) Entity (b) Constraints
- (c) Rules (d) Keys

Ans : (b) Constraints enforce limits to the data or type of data that can be inserted/updated/deleted from a table. The whole purpose of constraints is to maintain the data integrity during an update/delete/insert into a table.

15. Data Integrity control uses.....

- (a) Upper and lower limits on numeric data
- (b) Passwords to prohibit unauthorized access to files
- (c) Data dictionary to keep the data
- (d) Data dictionary to find last access of data

Ans : (b) Passwords to prohibit unauthorized access to because option (a) is defined by integrity constraints. And option (c) and (d) is not related to data integrity control.

16. What does the following declaration mean? int (xptr) [10]

- (a) ptr is an array of pointers of 10 integers
- (b) ptr is a pointer to an array of 10 integers
- (c) ptr is an array of 10 integers
- (d) none of the above

Ans : (b) int (* ptr) [10]. means ptr is a pointer to an array of 10 integers.

17. Which of the following has compilation error in C?

- (a) `int n = 32`
- (b) `char ch = 65`
- (c) `float f = (float) 3.2`
- (d) none of the above

Ans : (d) All are the valid declaration statement in C.

18. Which of the following operators cannot be overloaded in C++?

- (a) `?`
- (b) `+=`
- (c) `==`
- (d) `::`

Ans : (d) Following is the list of operators, which cannot be overloaded.

- (i) scope resolution operator (`::`) ?
- (ii) Member selection operator (`.`(dot), `.*`)
- (iii) Conditional operator `?:`
- (iv) `sizeof()` operator

These are the operators which cannot be overloaded, So in given option `::` cannot be over loaded.

19. Allows to create classed which are derived from other classes, so that they automatically include some of its "parent's" members, plus its own members.

- (a) Overloading
- (b) Inheritance
- (c) Polymorphism
- (d) Encapsulation

Ans : (b) Inheritance can be defined as the process where one class acquires the properties of another class. Inheritance allows to create classes which are derived from other classes, so that they automatically include of it parent's member plus it member.

20. The correct way to round off a floating number x to an integer value is

- (a) `y = (int) (x + 0.5)`
- (b) `y = int (x + 0.5)`
- (c) `y = (int) x + 0.5`
- (d) `y = (int) ((int) x + 0.5)`

Ans : (a) Lets take `x = 10.3`, so after round off it should be 10

and if `x = 10.5` after round off it should be 11 now option (a) $10.3 + 0.5 = 10.8 \Rightarrow \text{int}(10.8) = 10$ correct also $10.5 + 0.5 = 11 \Rightarrow \text{int}(11) = 11$ again correct

(b) incorrect syntax

(c) $y = (\text{int}) x + 0.5 \Rightarrow \text{int}(10.3) = 10 + 0.5 = \text{int}(10.5) = 10$ correct but $\text{int}(10.5) = 10 + 0.5 = 10.5 \Rightarrow \text{int}(10.5) = 10$ incorrect

(d) $\text{int}(\text{int}(10.5) + 0.5) = \text{int}(10 + 0.5) = \text{int}(10.5) = 10$ incorrect.

21. What is the value of the postfix expression? a b c d + - × (where a = 8, b = 4, c = 2 and d = 5)

- (a) $-3/8$
- (b) $-8/3$
- (c) 24
- (d) -24

Ans : (d) Postfix expression = $abcd + - \times$

$\Rightarrow ab (c + d) - \times$

$\Rightarrow a(b - (c + d)) \times$

$\Rightarrow (a \times (b - (c + d)))$

now put the values $a = 5, b = 4, c = 2, d = 5$

$= (8 \times (4 - (2 + 5)))$

$= (8 \times (4 - 7))$

$= (8 \times (-3))$

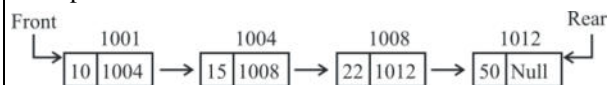
$= -24$

22. If the queue is implemented with a linked list, keeping track of a front pointer and a rear pointer, which of these pointers will change during an insertion into a non-empty queue?

- (a) Neither of the pointer change
- (b) Only front pointer changes
- (c) Only rear pointer changes
- (d) Both of the pointers changes

Ans : (c) In linked list implementation of queue the last inserted node is always pointed by 'rear' and first node is always pointed by 'front'.

Example :



In above example, the last inserted node is 50 & it is pointed by 'rear' and the first inserted node is 10 & it is pointed by 'front'. The order of elements inserted is 10, 15, 22 and 50 since the insertion is done from the rear end in queue therefore the rear pointer will change only.

23. is often used to prove the correctness of a recursive function.

- (a) Digitalization
- (b) Communitivity
- (c) Mathematical Induction
- (d) Matrix Multiplication

Ans : (c) It is a mathematical proof technique, It is essentially used to prove that a property $P(n)$ holds for all natural numbers 0, 1, 2, 3 & so on.

Correctness of recursive algorithms is after proven by mathematical induction.

24. For any B-tree of minimum degree $t \geq 2$, every node other than the root must have atleastkeys and every node can have at most keys.

- (a) $t - 1, 2t + 1$
- (b) $t + 1, 2t + 1$
- (c) $t - 1, 2t - 1$
- (d) $t + 1, 2t - 1$

Ans : (c) In B-tree every internal node other than the root is at least half-full,

i.e. $t - 1 \leq \# \text{ keys} \leq 2t - 1$

$t \leq \# \text{ children} \leq 2t$

25. Given two sorted list of size 'm' and 'n' respectively. The number of comparison needed in the worst case by the merge sort algorithm will be

- (a) $m \times n$ (b) $\max(m, n)$
(c) $\min(m, n)$ (d) $m + n - 1$

Ans : (d) The number of comparisons needed in the worst case by the merge sort algorithm will be $m + n - 1$ i.e. only last element will not be compare only.

26. Given the following statements :

S1 : SLR uses follow information to guide reductions. In case of LR and LALR parsers, the look-aheads are associated with the items and they make use of left context available to the parser.

S2: LR grammar is a larger subclass of context free grammar as compared to that SLR and LALR grammars.

Which of the following is true?

- (a) S1 is not correct and S2 is not correct
(b) S1 is not correct and S2 is correct
(c) S1 is correct and S2 is not correct
(d) S1 is correct and S2 is correct

Ans : (d) S1 is correct & S2 is correct.

SLR parser uses the follow information to construct the SLR parser table. LR and LALR uses the look-ahead symbols to generate the parse table-LR is powerful than SLR and LALR. Therefore, LR will be able to recognize large class of CFG than the SLR and LALR.

27. The context free grammar for the language $L = \{a^m b^n \mid n \leq m + 3, n \geq 0, m \geq 0\}$ is

- (a) $S \rightarrow aaa A; A \rightarrow aAB \mid B, B \rightarrow Bb \mid \lambda$
(b) $S \rightarrow aaaA \mid \lambda A \rightarrow aAb \mid B, B \rightarrow Bb \mid \lambda$
(c) $S \rightarrow aaaA \mid aaA \mid \lambda A \rightarrow aAb \mid B, B \rightarrow Bb \mid \lambda$
(d) $S \rightarrow aaaA \mid aaA \mid aA \mid \lambda A \rightarrow aAb \mid B, B \rightarrow Bb \mid \lambda$

Ans : (b) $L = \{a^n b^m \mid n \leq m + 3; n \geq 0, m \geq 0\}$

The grammar will be,

$S \rightarrow aaaA \mid \lambda$

$A \rightarrow aAb \mid B$

$B \rightarrow Bb \mid \lambda$

28. Given the following statements :

S1 : If L is a regular language then the language $\{uv \mid u \in L, v \in L^R\}$ is also regular.

S2 : $L = \{ww^R\}$ is regular language.

Which of the following is true?

- (a) S1 is not correct and S2 is not correct
(b) S1 is not correct and S2 is correct
(c) S1 is correct and S2 is not correct
(d) S1 is correct and S2 is correct

Ans : (c) S1 is correct & S2 is incorrect

- If L is regular then reversed of L is also regular, hence U & V are regular
- Concatenation of regular languages are regular
- Hence UV is regular

S1 is correct

- To find ww^R need to remember what is w.
- Regular grammar is checked by finite automata, which has no memory (stack)
- To find ww^R we need at least NPDA

S2 is incorrect.

29. The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called.....

- (a) Symbol resolution (b) Parsing
(c) Assembly (d) Relocation

Ans : (d) Relocation is the process of replacing symbolic references or names of libraries with actual usable address in memory before running a program.

- It is typically done by the linker during compilation

It can be done at runtime by a relocating loader

Compilers or assemblers typically generate the executable with zero as the lower most starting address. Before the execution of object code, these address should be adjusted so that they denote the correct runtime address.

30. Which of the following derivations does a top-down parser use while parsing an input string? The input is scanned from left to right.

- (a) Leftmost derivation
(b) Leftmost derivation traced out in reverse
(c) Rightmost derivation traced out in reverse
(d) Rightmost derivation

Ans : (a) Top – down parser – leftmost derivation
bottom up parser – reverse of rightmost derivation

A top down parser is called LL parser because it parser the input from left to right and constructs a leftmost derivation.

31. The dual of a Boolean expression is obtained by interchanging

- (a) Boolean sums and Boolean products
(b) Boolean sums and Boolean products or interchanging 0's and 1's
(c) Boolean sums and Boolean products and interchanging 0's & 1's
(d) Interchanging 0's and 1's

Ans : (c) The **dual of a boolean expression** is obtained by interchanging boolean sums and boolean products as well as interchanging 0's as well as 1's.

32. Given that $(292)_{10} = (1204)_x$ in some number system x . The base x of that number system is :

- (a) 2 (b) 8
(c) 10 (d) None of these

Ans : (d) $(292)_{10} = (1204)_x$
 $2 \times 10^2 + 9 \times 10 + 2 = x^3 + 2 \times x^2 + 0 + 4$
 $200 + 90 + 2 = x^3 + 2x^2 + 4$
 $x^3 + 2x^2 - 288 = 0$
 Solution for $x = 6$
 none of the option satisfy the solution.

33. The sum products expansion for the function $F(x, y, z) = (x + y)\bar{z}$ is given as

- (a) $\bar{x}\bar{y}z + xy\bar{z} + \bar{x}y\bar{z}$ (b) $xyz + xy\bar{z} + x\bar{y}\bar{z}$
 (c) $x\bar{y}\bar{z} + \bar{x}y\bar{z} + xy\bar{z}$ (d) $xy\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$

Ans : (d) Use boolean identities to expand the product and simplify:

$$\begin{aligned} f &= (x + y) z' \\ &= xz' + yz' \\ &= xz' (1) + yz' (1) \\ &= xz' (y + y') + yz' (x + x') \\ &= xyz' + xy'z' + xyz' + x'yz' \\ &= xyz' + xy'z' + xyz' \end{aligned}$$

34. Let $P(m, n)$ be the statement "m divides n" where the universe of discourse for both the variables is the set of positive integers. Determine the truth values of each of the following propositions:

- (i) $\forall m \forall n P(m, n)$
 (ii) $\exists m \forall n P(m, n)$
 (a) Both (i) and (ii) are true
 (b) Both (i) and (ii) are false
 (c) (i) false & (ii) true
 (d) (i) true & (ii) false

Ans : (c) $P(3, 2)$ is True because 3 does not divide 2. So only II statement is true.

35. Big – O estimate for $f(x) = (x + 1) \log(x^2 + 1) + 3x^2$ is given as

- (a) $O(x \log x)$ (b) $O(x^2)$
 (c) $O(x^3)$ (d) $O(x^2 \log x)$

Ans : (b) Big O notation of $f(x) = (x + 1) \log(x^2 + 1) + 3x^2$

Note. $(x + 1)$ is $O(x)$ and $x^2 + 1 \leq 2x^2$. When $x > 1$

So,

$$\log(x^2 + 1) < d = \log(2x^2) = \log 2 + \log x^2 = \log 2 + 2 \log x \leq 3 \log x \text{ if } x^2$$

Thus, $\log(x^2 + 1)$ is $O(x \log x)$

Also, $3x^2$ is $O(x^2)$

So, $f(x)$ is $O(\max(x \log x, x^2)) = O(x^2)$ as

$x \leq x^2 \log x$ for x .

36. How many edges are there in a forest of t -trees containing a total of n -vertices?

- (a) $n + t$ (b) $n - t$
(c) $n * t$ (d) n^t

Ans : (b) In each tree we have $K-1$ edge for K vertices, so, for t trees with total n vertices (all trees are disconnected in a forest) we would have $n - t$ edges

37. Let f and g be the functions from the set of integers to the set integers defined by

$$f(x) = 2x + 3 \text{ and } g(x) = 3x + 2$$

Then the composition of f and g and g and f is given as

- (a) $6x + 7, 6x + 11$
(b) $6x + 11, 6x + 7$
(c) $5x + 5, 5x + 5$
(d) None of these

Ans : (a) Given $f(x) = 2x + 3$

$$g(x) = 3x + 2$$

$$(f \circ g)(x) = f(g(x))$$

$$f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

$$(g \circ f)(x) = g(f(x))$$

$$g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

38. If n and r are non-negative integers and $n \geq r$, then $p(n+1, r)$ equals to:

- $$\begin{array}{ll} \text{(a)} & \frac{P(n,r)(n+1)}{(n+1-r)} \\ \text{(c)} & \frac{P(n,r)(n+1)}{(n+1-r)} \end{array} \quad \begin{array}{ll} \text{(b)} & \frac{P(n,r)(n+1)}{(n-1+r)} \\ \text{(d)} & \frac{P(n,r)(n+1)}{(n+1+r)} \end{array}$$

Ans : (a) $P(n, r) = \frac{n!}{(n-r)!}$

$$P(n+1, r) = \frac{(n+1)!}{(n+1-r)!} = \frac{(n+1) \times n!}{(n+1-r) \times (n-r)!}$$

$$= \frac{(n+1)}{(n+1-r)} \times P(n, r)$$

39. A graph is non-planar if and only if it contains a sub graph homomorphism to :

- (a) $K_{3,2}$ or K_5
 (b) $K_{3,3}$ and K_6
 (c) $K_{3,3}$ or K_5
 (d) $K_{2,3}$ and K_5

Ans : (c) According to Kuratowski's theorem a finite graph is planer is and only if it does not contain a sub graph that is a sub division of K_5 (the complete graph on 5 vertices) or of $K_{3,3}$ (complete bipartite graph on six vertices, three of which connect to each of the other three, also known as utility graph)

Wireless data gram protocol is protocol in WAP architecture, covers the transport layer, protocol in the internet model 8 UDP is used for wireless Datagram protocol, so it satisfies (iv)
wireless is a internet protocol (IP)

43. Which of the following is widely used inside the telephone system for long-haul data traffic?

- (a) ISDN
- (b) ATM
- (c) Frame Relay
- (d) ISTN

Ans : (b) ATM is a core protocol used over the sonet/SDH backbone of the public switched telephone network and ISDN.

44. The document standards for EDI were first developed by large business house during the 1970s and are now under the control of the following standard organization :

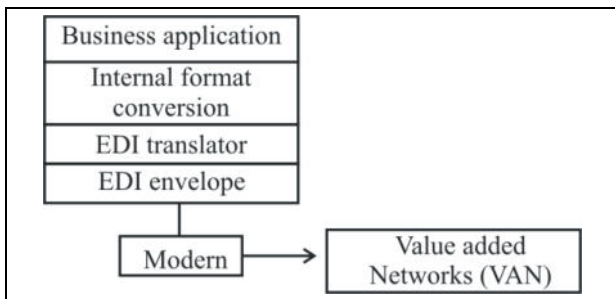
- (a) ISO
- (b) ANSI
- (c) ITU-T
- (d) IEEE

Ans : (b) EDI service provides maintain a VAN and establishes mail boxes for each business partner involved in EDI. The provider stores and forwards EDI message between the partners. The main aspect here is standardization. All the parties involved in EDI transaction must use an agreed set of document layout standards. The document stores for EDI were developed by large business houses during the 1970's and are now under the control of the American national standard institute (ANSI)

45. Electronic Data Interchange Software consists of the following four layers :

- (a) Business application, Internal format conversion, Network translator, EDI envelope
- (b) Business application, Internal format conversion, EDI translator, EDI envelope
- (c) Application layer, Transport layer, EDI translator, EDI envelope
- (d) Application layer, Transport layer, IP layer, EDI envelope

Ans : (b) EDI consist of four layers : Business application, internal format conversion, EDI translator, EDI envelope



46. Consider a pre-emptive priority based scheduling algorithm based on dynamically changing priority. Larger priority number implies higher priority. When the process is waiting for CPU in the ready queue (but not yet started execution), its priority changes at a assigned priority value or when they enter ready queue. Assume that the following processes want to execute : Process ID Arrival Time Service Time P1 0 4 P2 1 1 P3 2 2 P4 3 1. The time quantum $q = 1$. When two processes want to join ready queue simultaneously the process which has not executed recently is given priority. The finish time of processes P1, P2, P3 and P4 will respectively be :

- (a) 4, 5 and 8 (b) 8, 2 7 and 5
(c) 2, 5, 7 and 8 (d) 8, 2, 5 and 7

Ans : (b)

Following processes want to execute :

Process ID	Arrival time	Service time
P1	0	4
P2	1	1
P3	2	2
P4	3	1

Time	Priority
0	P1 = 1
1	P2 = 1, P1 = 2
2	P2 = 2, P1 = 2, P3 = 0
3	P1 = 1, P3 = 2, P4 = 0
4	P3 = 1, P1 = 2, P4 = 4
5	P1 = 1, P3 = 2, P4 = 2

Gantt chart

0 – P1 – 1 – P2 – 2 – P1 – 3, – P3 – 4 – P4 – 5 – P1 – 6 – P – 7 – P1 – 8

P1 finishes at 8

P2 finishes at 2

P3 finishes at 7

P4 finishes at 5

So answer is (b)

47. The virtual address generated by a CPU is 32 bits. The translation look-aside buffer (TLB) can hold total 64 page table entries and a 4-way set associative (i.e. with 4-cache lines in the set).

The page size is 4 KB. The minimum size of TLB tag is :

- (a) 12 bits
- (b) 15 bits
- (c) 16 bits
- (d) 20 bits

Ans : (c) Page size of 4 KB, so offset bits are 12 bits.

Remaining bits of virtual address $32 - 12 = 20$ bits will be used for indexing.

Number of sets = $64/4 = 16$ (4 – way set) $\Rightarrow 4$ bits

So tag bits = $20 - 4 = 16$ bits

48. Consider a disk queue with request for input/output to block on cylinders 98, 183, 37, 122, 14, 124, 65, 67 in that order. Assume that disk head is initially positioned at cylinder 53 and moving towards cylinder number 0. The total number of head movements using shortest seek time first (SSTF) and SCAN algorithms are respectively :

- (a) 236 and 252 cylinders
- (b) 640 and 236 cylinders
- (c) 235 and 640 cylinders
- (d) 235 and 252 cylinders

Ans : No the these 236 and 252 cylinders According to SCAN 53 to 37 then to 14 then to 0 ($16 + 23 + 14 = 53$) now direction will be reversed so 0 0 65 then to 67 then to 99, 122, 124, 183 ($65 + 2 + 31 \rightarrow 24 + 2 + 59 = 183$) total head movements = $183 + 53 = 236$

According to SSTE 53 to 65 then to 67 then 37m 14, 98, 122, 124, 183 (so head movements = $12 + 2 + 30 + 23 + 84 + 24 + 59 = 236$)

49. How much space will be required to store the bit map of a 1.3 GB disk with 512 bytes block size ?

- (a) 332.8 KB
- (b) 83.6 KB
- (c) 266.2 KB
- (d) 256.6 KB

Ans : (a) 332.8 KB

We need a bit for each block. Number of block = disk size/block size = $1.3 \text{ GB} / 512 \text{ MB}$

This will give the size of bit map in bits. 70 get the byte size divide by 8 and we get 332.8 KB. For disk size, GB is taken as $1024 \times 1024 \times 1024$ bytes & KB it is 1024 bytes

So answer is (a)

50. Linux operating system uses :

- (a) Affinity scheduling
- (b) Fair preemptive scheduling
- (c) Hand shaking
- (d) Highest penalty ratio next

Ans : (b) (i) A time-sharing algorithm for fair preemptive scheduling between multiple processes.

(ii) A real-time algorithm for tasks where absolute priorities are more important than fairness.